

Niramay M. Patel

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Education

Indian Institute of Science Education and Research (IISER), Bhopal
BS in Engineering Science (Major in Computer Science)

August 2023 – Present

- CPI: 9.33/10.0

Atmiya Vidya Mandir, Surat, Gujarat

Passed in March 2023

- CBSE Std. 12 Result: 94.8%

Experience

Research Intern (On site)

VisDom Lab, IISERB

Professor: Dr. Vinod Kurmi

May 2025 – Present

- Conducted research on **fairness in deep learning**, including debiasing with bias ranking, fairness analysis of the Descript Audio Codec (DAC), and fairness-aware coreset selection.
- Proposed **Evidence-Gated Suppression (EGS)**, a regularizer that tracks class-conditional, confidence-weighted neurons evidence and applies percentile-gated decay to suppress them during training, improving worst-group robustness without requiring group labels.
- Developed a **reweighting framework** that combines per-example hardness and group-level loss signals to improve robustness and fairness in dermatology image classification across Fitzpatrick skin types.

Research Intern (Remote)

IIT Roorkee, India

Prof: Dr. Raksha Sharma

Jan 2025 – Present

- Conducted research to improve the robustness of **speech-to-text models** under real-world noise by designing a controllable procedural noise simulator and analyzing low-dimensional environmental structure via PSD-based spectral representations.
- Performed large-scale robustness experiments using real and simulated noise, studying noise-injected training and evaluation across clean, real-world, and synthetic acoustic conditions.
- Designed a structured **hyperparameter tuning framework** and proposed a principled evaluation protocol to quantify performance stability and robustness across diverse acoustic environments.

Research Intern (On site)

MOON Lab, IISERB

Prof: Dr. P. B. Sujit

Dec 2024 – Jan 2025

- Studied core topics in **Computer Vision**, including OpenVINS, MSKCF, SLAM algorithms, and the mathematical foundations of sensor fusion.
- Implemented **Visual Odometry** on the KITTI dataset, building strong theoretical and practical expertise.

Publications

- **Regulating Internal Evidence Flows for Robust Learning Under Spurious Correlations**
Accepted at ICLR 2026 ([OpenReview](#) 📄)
- **Robustness to Skin Type Using Fused Upweighting**
Submitted to MICCAI 2026
- **Hyperparameter Strategies for Noise-Robust Speech-to-Text**
Submitted to ACL ARR 2026

Projects

Seeing the Object, Not the Context

course project, October 2025

- Addressed the problem of **spurious background correlations** in deep vision models that degrade generalization under distribution shift.

- Developed a training-time method that **redirects model attention to foreground objects**, effectively neutralizing background bias without requiring extra supervision.
- Achieved consistent **10–15% absolute improvements in accuracy and F1-score** across bias-sensitive benchmarks, with strong robustness under conflicting background conditions and no inference-time modifications.

Fairness in DAC

[github repo](#) 

VisDom Lab, June 2025

- Analyzed potential bias in **ParlerTTS** by studying **DAC RVQ codebooks** to examine whether gender-specific patterns appear at the acoustic token level.
- Designed an inspection pipeline to **capture RVQ tokens before decoding** and visualized gender-swapped prompts using **t-SNE**, finding largely uniform token distributions that suggest bias originates upstream rather than in DAC.

Spuriousity Ranking Stability

VisDom Lab, May 2025

- Conducted a robustness study for **Spuriousity Ranking Stability** by analyzing how SEBRA's self-guided bias ranking behaves under noisy labels, random initialization, and dataloader stochasticity, revealing key reproducibility limitations.
- Systematically evaluated ranking consistency and accuracy trade-offs, showing that **data sampling order dominates ranking variance** while the core SEBRA mechanism remains theoretically sound but practically fragile.

Visual Odometry

[github repo](#) 

MOON LAB, December 2024

- Built strong theoretical grounding through extensive reading on **quaternion geometry**, Kalman Filter, error-state modeling, and drone kinematics.
- Estimated the camera's motion from a sequence of images, a crucial task in autonomous driving and SLAM. Successfully reconstructed the **vehicle's trajectory** and evaluated the accuracy using ground truth data.

Certifications

Machine Learning Specialization by Stanford University and DeepLearning.AI

[certificate](#) 

- **Topics covered:** Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommenders, and Reinforcement Learning.

Co-Curricular and Extra-Curricular

- Proposed an **AI based Dynamic Bus route rationalization** model that would be required based on real-time traffic and road parameters for the **Delhi Government** in Smart India Hackathon.
- Secured a rank **under 500** as a College Ambassador for **Techfest, IIT Bombay**.
- Led the **core committee** in organizing a two-day Navaratri event attended by over 1,000 participants, managing end-to-end planning and execution.
- I enjoy playing table tennis, badminton, and cricket to relax during my free time.

Research Interests

My research interests center on **bias and fairness in Artificial Intelligence**, with a focus on principled analysis of modern learning systems. I have indepth working knowledge of **Reinforcement Learning**. Moreover, I have strong experience working with **speech and audio models**, particularly generative AI models. I am motivated to pursue research across core areas of artificial intelligence.